SMART INDIA HACKATHON 2026

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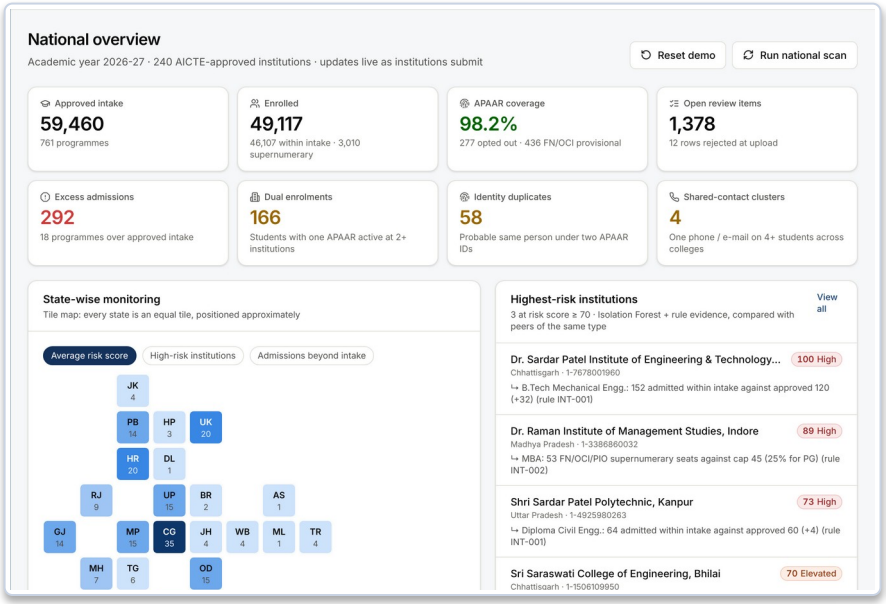
TITLE PAGE

- **Problem Statement ID –** SIH26263
- **Problem Statement Title-** Development of an APAAR ID-Based Student Data Management and Verification Platform for AICTE-Approved Institutions
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-** 194599
- **Team Name (Registered on portal)** AI Riders

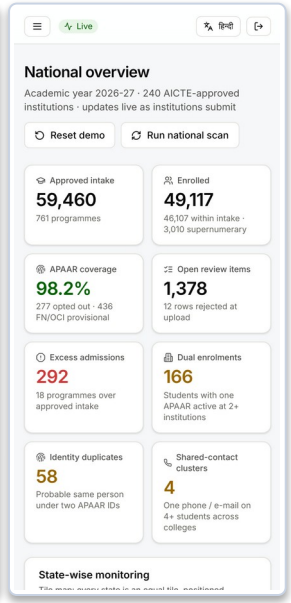
VIDYA PRAMAAN

One student. One APAAR ID. One verifiable record.
Live: vidya-pramaan-sih26263.onrender.com
Demo video (5 min): youtu.be/eBwySqMs63Q
7 roles · 31 rules · 65 tests · synthetic data

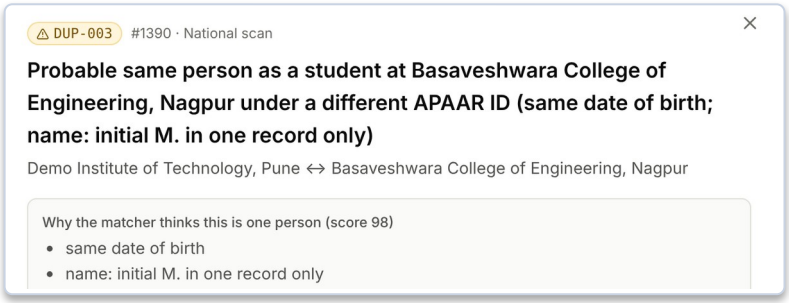
Working prototype · live now, every screen below is real



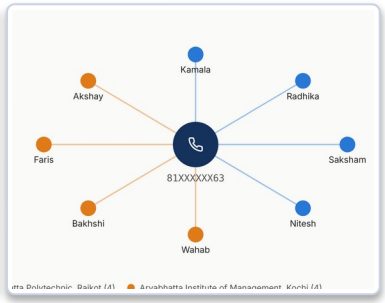
AICTE national view: every breach named with its rule



On a phone (PWA)



Same person under two APAAR IDs: reasons shown



1 phone, 8 students, 2 colleges

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VIDYA PRAMAAN

29.6 lakh enrolments in AICTE institutions (2025-26) · **24.25 crore** APAAR IDs issued (Mar 2026)

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❖ Proposed Solution (Describe your Idea/Solution/Prototype)

❖ Detailed explanation of the proposed solution



Institution upload
Excel · CSV · ERP API

APAAR validation
registry + consent

Data verification
31 explained rules

Central repository
one record per student

Analytics & monitoring
after every submission

Authorised verification
consent + signed QR

What is new no national system checks each admission against approved intake. We add that check, on top of APAAR.

❖ How it addresses the problem

✖ AICTE gets only totals (PS)	✔ Student-level records on APAAR
✖ Excess seat: 5× fee fine each (APH)	✔ Seat #61 of 60 refused at upload
✖ Same student at two colleges	✔ Cross-college match + contact rings
✖ Hard to verify at source (PS)	✔ Signed QR, checks offline

❖ Innovation and uniqueness of the solution

Twin-safe matching
Mohd = Mohammed
Rohit ≠ Mohit

Ghost-admission map
1 phone, 8 students, 2 colleges

Decisions kept
with a reason; never re-flagged

Signed checkpoints
history rewrites detected

Rules as data
sourced; unconfirmed values only warn

APAAR on degrees
university print feed + QR

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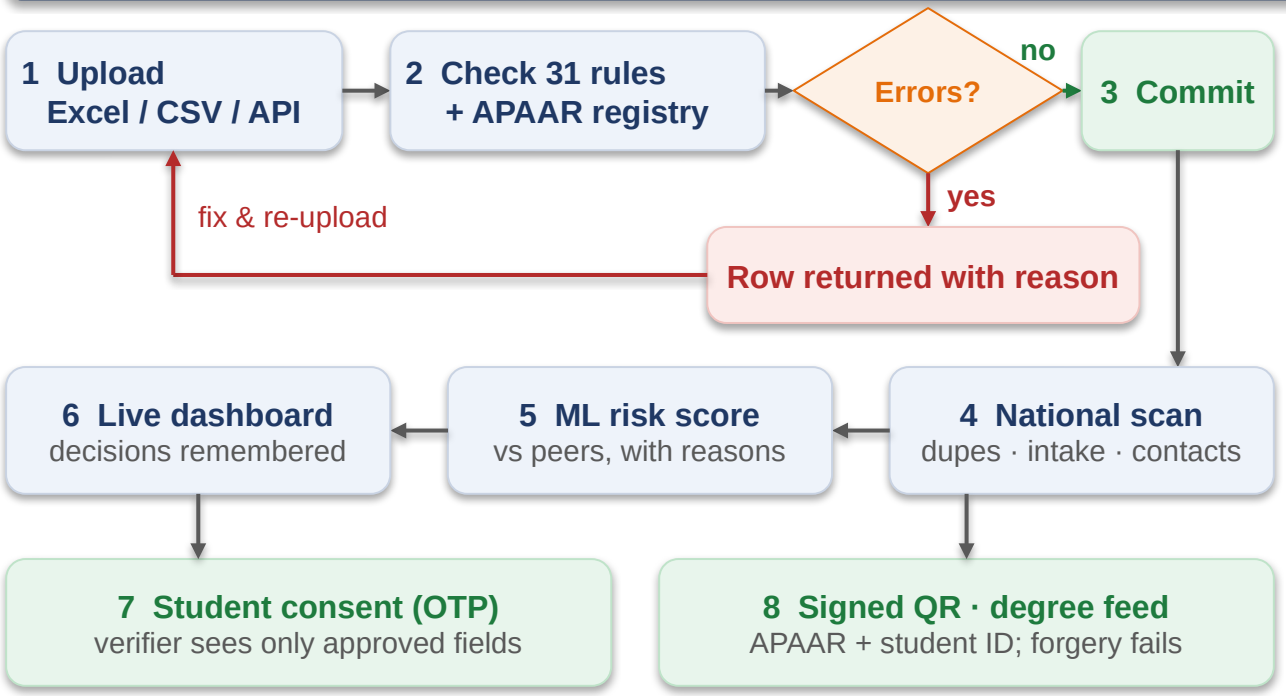
TECHNICAL APPROACH

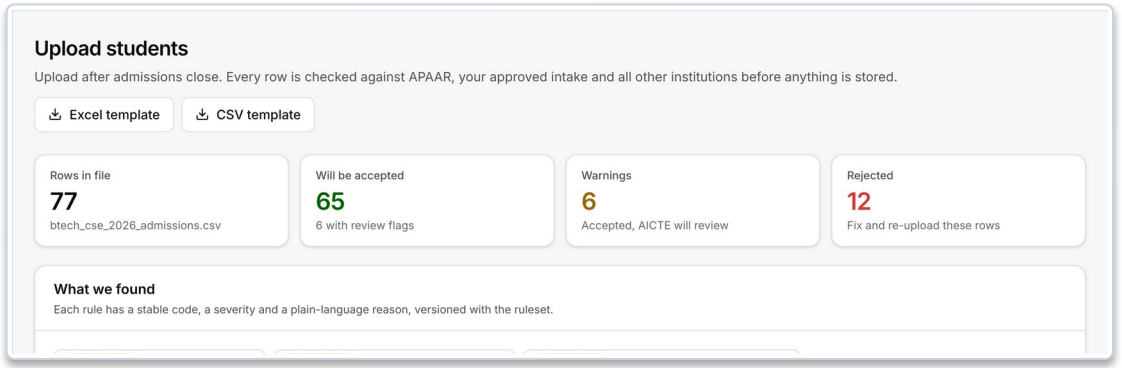
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❖ Technologies to be used (e.g. programming languages, frameworks, hardware)

- **Python · FastAPI** — REST · OpenAPI
- **React · TypeScript** — installable PWA
- **PostgreSQL 16** — SQLite in the demo
- **scikit-learn** — Isolation Forest
- **Name matcher** — twin-safe · Indic
- **Ed25519 · W3C** — did:web · AES-GCM

❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)





Working prototype: 77-row upload → 65 accepted, 12 returned with rule codes

Built: upload, 31 rules, scan, risk score, consent, signed QR, audit

Simulated (no public API yet): APAAR lookup, DigiLocker, portal sign-in

Planned: API Setu / DigiLocker partner access, govt SSO, STQC audit

Try it live: https://vidya-pramaan-sih26263.onrender.com

Demo video (5 min): youtu.be/eBwySqMs63Q · 7 roles · synthetic data

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FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea

0.79 / 0.92
name-match precision / recall on 49 held-out pairs (old 0.65 / 0.68)

0 of 80
seeded twin pairs wrongly merged (held-out: 2 of 12, sent to an officer)

17 of 17
error types caught in the demo upload; 0 false flags on 56 clean rows

225 of 225
planted cases found: 164 dual enrolments, 57 ID duplicates, 4 contact rings

42 of 46
highest-risk colleges were the planted anomalies

65 tests
93% line coverage; security and WCAG checks in CI

All synthetic data: regression checks, not real-world accuracy (the pilot measures that). An officer decides every flag.

Potential challenges and risks	Strategies for overcoming these challenges
No public APAAR API	✔ Adapter: mock now → DigiLocker / API Setu
Students without APAAR (opt-out, FN / OCI)	✔ Provisional ID; never rejected
Privacy (DPDP Act; APAAR is voluntary)	✔ Field-level consent; contacts encrypted; no Aadhaar
Wrong flags: twins, spellings, bad data	✔ Reasons shown; officer decides; decision kept
Rules change; some values unconfirmed	✔ Rules file per year; unconfirmed values only warn
Varied ERP formats, low digital readiness	✔ Headers auto-mapped; Excel template; Hindi; phone

Roadmap M1–3 pilot: 1 state, 50 colleges, measure hours saved → M4–6 API Setu, SSO, STQC → M7–12 national **Cost** no licence fees

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IMPACT AND BENEFITS

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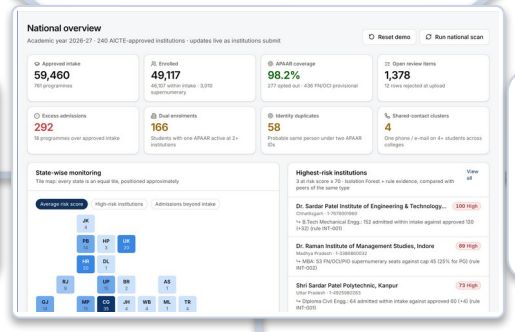
❖ Potential impact on the target audience

AICTE & regulators
excess and dual enrolment at submission

Institutions
one upload; errors explained

Universities
APAAR printed on degrees

Employers
QR check; consent for more



One national view

Students
control who sees their record

Today
✖ ₹23.1 cr fine: 42 seats × ₹11 lakh × 5 (SC, 2019)

✖ AISHE 2022-23 data out in 2026

✖ APAAR–Aadhaar mismatches (CU Srinagar, 2026)

With Vidya Pramaan
✔ Seat #61 of 60 refused at upload

✔ National view after each submission

✔ Mismatch returned with its reason

❖ Benefits of the solution (social, economic, environmental, etc.)

Social
APAAR opt-outs never refused (SC, 2026)

Economic
no licence fees; one upload serves AICTE + university

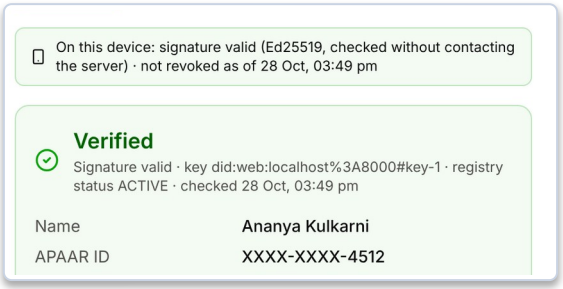
Environmental
fewer paper returns and letters

Governance
audited evidence pack for show-cause

Expected benefits; the pilot measures them.



Certificate: APAAR ID + student ID + QR



QR: on-device check + revocation list

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RESEARCH AND REFERENCES

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❖ Details / Links of the reference and research work

Why not just use the systems that exist?

	NAD / DigiLocker	ABC	AISHE	Vidya Pramaan
Student-level admissions vs approved intake	—	—	totals, 2–3 yrs late	✔ **per upload**
Same student at two colleges; ghost admissions	—	—	—	✔
Supernumerary caps by programme and year	—	—	—	✔
Consent-based verification for employers	✔	—	—	✔ **field-level**
APAAR + student ID feed for degree printing	issuer uploads	credits only	—	✔

✔ **= built and tested (docs/CLAIMS.md).** Reuses APAAR; no new ID.

References

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4. NAD on DigiLocker: consent-based verification (FAQ) nad.digilocker.gov.in
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9. W3C Verifiable Credentials and did:web; RFC 8032 (Ed25519) w3.org · rfc-editor.org
10. Liu, Ting & Zhou (2008) Isolation Forest; IDinsight on matching Indian names doi.org/10.1109/ICDM.2008.17 · idinsight.org

Demo video, 5 min
youtu.be/eBwySqMs63Q
every role, narrated, captioned

Live prototype
vidya-pramaan-sih26263.onrender.com
7 demo roles · synthetic data

Source code & tests
github.com/Pulkitkh/VidyaPramaan
65 tests · 93% coverage · MIT